

SANA ULLAH

M.Phil. Biotechnology | AI in Agriculture | Molecular Genetics sanaullah2282@gmail.com
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PERSONAL STATEMENT

Results-driven Biotechnology researcher with an M.Phil. (CGPA 3.77/4.00) specializing in AI-assisted agricultural diagnostics, molecular genetics, and plant biotechnology. Experienced in designing and executing laboratory-based research, field data collection, and integrating machine learning techniques with biological sciences. Seeking research, academic, or industry positions where cutting-edge biotechnology and artificial intelligence can be applied to solve real-world agricultural and biomedical challenges.

EDUCATION

Institution	Degree	Session	CGPA / Marks
University of Agriculture Peshawar, KPK, Pakistan	M.Phil. Biotechnology	2022 – 2026	3.77 / 4.00
University of Agriculture Peshawar, KPK, Pakistan	B.Sc. (Hons) Biotechnology	2018 – 2022	3.60 / 4.00
Government College Peshawar, KPK, Pakistan	F.Sc. Pre-Medical	2014 – 2016	820 / 1100
Frontier Grammar School, Ghari Sherdad, KPK, Pakistan	Matriculation (Science)	2012 – 2014	966 / 1100

RESEARCH INTERESTS

- Molecular Genetics and Genomics
- Recombinant DNA Technology
- Artificial Intelligence in Biotechnology and Precision Agriculture
- Bioinformatics and Computational Biology

RESEARCH EXPERIENCE

M.Phil. Thesis Research

2022 – 2026

University of Agriculture Peshawar, KPK, Pakistan

Artificial Intelligence-Assisted Real-Time Detection of Devastating Fruit Fly Species for Population Monitoring and Effective Control

- Developed and validated a deep learning-based detection system for economically damaging fruit fly species
- Integrated computer vision and real-time image processing pipelines for accurate pest identification in field conditions
- Contributed to evidence-based pest control strategies through population monitoring data

Research Internship

2021 – 2022

Plant Tissue Culture & Germplasm Conservation Lab, Institute of Biotechnology and Genetic Engineering, University of Agriculture Peshawar

Project: Agromorphological and Biochemical Characterization of Advanced Wheat Genotypes under Field Conditions

- Conducted agromorphological assessments of elite wheat genotypes under varying field environments
- Performed biochemical assays to evaluate genotype performance and stress tolerance indicators
- Contributed to germplasm conservation protocols and data documentation

WORK EXPERIENCE

Field Officer

Feb 2021 – Feb 2024

Dawn Agro Services, Peshawar, Pakistan

- Conducted regular field visits and monitored agricultural practices across 20+ farm sites
- Advised farmers on evidence-based crop management, integrated pest control, and soil health strategies
- Collected, recorded, and analyzed field performance data to track crop health and yield outcomes
- Maintained effective coordination between farming communities and company management
- Demonstrated consistent professionalism, reliability, and strong work ethic in field operations

LABORATORY & TECHNICAL SKILLS

Molecular Biology: Plant DNA Extraction, PCR, Gel Electrophoresis, CTAB Preparation, Nanodrop Spectrophotometry, Centrifugation

Bioinformatics: Sequence analysis, literature database navigation (NCBI, PubMed) **AI & Computing:** Computer Vision, Deep Learning (applied), MS Office Suite

Field Research: Agromorphological assessment, biochemical characterization, crop monitoring **General**

Skills: Scientific writing, data interpretation, research communication

CERTIFICATIONS & TRAINING

Certificate Course in Applied Biomedical AI

July – August 2025

OIC-COMSTECH in collaboration with Ningbo University, China

HONORS & AWARDS

- Awarded trophy and shield for achieving 3rd position in SSC examination (2014) — district-level recognition
- Received top academic awards consecutively from Class 1 through Class 8 for ranking 1st in annual examinations

LANGUAGES

Urdu: Native **Pashto:** Native

English:

Professional Working Proficiency

REFERENCES

Dr. Muhammad Sayyar Khan (Ph.D., Germany)

Associate Professor, Institute of Biotechnology and Genetic Engineering (IBGE)

University of Agriculture Peshawar, KPK, Pakistan

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M.Phil. Thesis Supervisor

Dr. Quahir Sohail

Associate Professor, Biochemistry and Plant Biotechnology Division

Institute of Biotechnology and Genetic Engineering (IBGE)
University of Agriculture Peshawar, KPK, Pakistan

Email:

Academic Referee

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